Stage-wise Extreme Atomic Proximity in HPacker Side-chain Reconstruction

Fixed-archive technical note | 24 October 2026 | Sho Miyazaki · Independent researcher, Tokyo, Japan

Abstract

We examined extreme interresidue heavy-atom proximity across six sequential stages of HPacker side-chain reconstruction applied to 200 archived frames from a single BioEmu wild-type ensemble. At stage 5, 134/200 frames had at least one nonadjacent interresidue heavy-atom distance strictly below 1.6 Å. PHE6–PHE55 exhibited ordered contact–separation–recontact in 43/200 frames, with complete six-stage alternation in 25/200. Selected, post hoc local interventions elicited structural and neural-network responses. In a separately identified continuation, four selected baseline/intervention-derived geometries completed short BioEmu relaxation, and a restricted protein heavy-atom Lennard-Jones calculation identified severe local overlap as the dominant source of its extremely large pre-relaxation repulsion. These fixed-input case results do not demonstrate general packing improvement, thermodynamic stabilization or increased MD success rates.

1. Introduction

Iterative side-chain reconstruction repeatedly updates the structural inputs used by subsequent reconstruction stages. Final-stage quality statistics alone do not show whether extreme atomic proximity appears, disappears and returns. HPacker already incorporates input-dependent side-chain modeling and repeated refinement [2]. Existing work has evaluated atomic clashes and energetic quality in protein backmapping [3]. The present fixed-input record focuses on stage-wise contact histories and selected local-response diagnostics, rather than a new packing algorithm.

2. Methods

2.1. Fixed ensemble and endpoints

The archive comprises 200 source frames (IDs 0–199) from one 63-residue BioEmu WT ensemble, each processed through HPacker stages 0–5 (1,200 archived stage structures). HPacker commit: 6cd5ec33961bb2662b51c0a15d11d6fd9162f2a7. Stage denotes reconstruction iteration, not MD time. The general close-contact endpoint counts a frame with any nonadjacent interresidue heavy-atom pair at distance strictly below 1.6 Å. PHE6–PHE55 uses the minimum of their interresidue heavy-atom distances. Recontact requires an ordered contact–absence–contact triple; complete alternation is exactly 010101 or 101010 (0: apart; 1: contact). These diagnostics are not MolProbity clashscore [4].

2.2. Exploratory local intervention and response tracing

In the corrected v8.1/v8.2 intervention implementation, side-chain masking excluded non-backbone atoms of the selected residue from the neighborhood arrays res_ids, atom_names, elements, coords and charges, without removing the residue from the PDB. A two-stage lag temporarily substituted side-chain coordinates saved at reconstruction stage $s-2$ during neighborhood evaluation and then restored the original coordinates; this branch was ineligible for s less than 2. Mediator χ substitutions were applied before Cartesian side-chain reconstruction. Earlier output-only edits did not constitute valid structural interventions. The saved v8.3/v8.4 diagnostics quantified Zernikegram, eight-component raw NN sin/cos output, circular χ and structural differences. High/low response labels are post hoc selected examples, not prospectively thresholded categories. Archived scripts and evidence tables in the repository identify conditions; complete end-to-end replay has not been independently certified.

2.3. Selected structural continuation and force-field settings

WT0 and WT1 baseline and v8.1 intervention-derived stage-5 structures (operationally termed “rescue” for local geometric change) underwent the BioEmu relaxation workflow. At BioEmu commit 42c551133f9825f4ec01812075b0d50e268c2209, OpenMM 8.2.0 with AMBER99SB (amber99sb.xml) and explicit TIP3P water (tip3p.xml) used 1.0-nm solvent padding, 0.1 M NaCl, PME with 1.0-nm nonbonded cutoff, constrained hydrogen bonds and rigid water. Langevin integration ran at 300 K with a final 1-fs step; four initial steps (10^{-6}, 10^{-5}, 10^{-4}, 10^{-3} ps) each covered 0.1 ps, followed after minimization by 0.1 ns NVT (10 ps^{-1} friction) and 0.4 ns NPT (1 ps^{-1}, Monte Carlo barostat at 1 bar). N, CA, C and O were position restrained by the original external restraint potential. Each selected run completed 611,100 integration steps; there was no production MD (simtime_ns=0). The saved execution used OpenCL and instrumented stepping. The separate Stage A evaluation considered only selected protein heavy-atom LJ pairs parameterized with AMBER99SB—not a solvated-system energy decomposition.

3. Results

3.1. Stage-wise contact prevalence

Across stages 0–5, general-contact frame counts were 183, 148, 140, 128, 130 and 134; 196/200 frames showed a general contact at least once. PHE6–PHE55 contact counts were 39, 22, 25, 23, 29 and 21. Across the six stages, 67/200 frames exhibited at least one PHE6–PHE55 contact, 43/200 showed ordered recontact and 25/200 showed full alternation (10 of 010101; 15 of 101010).

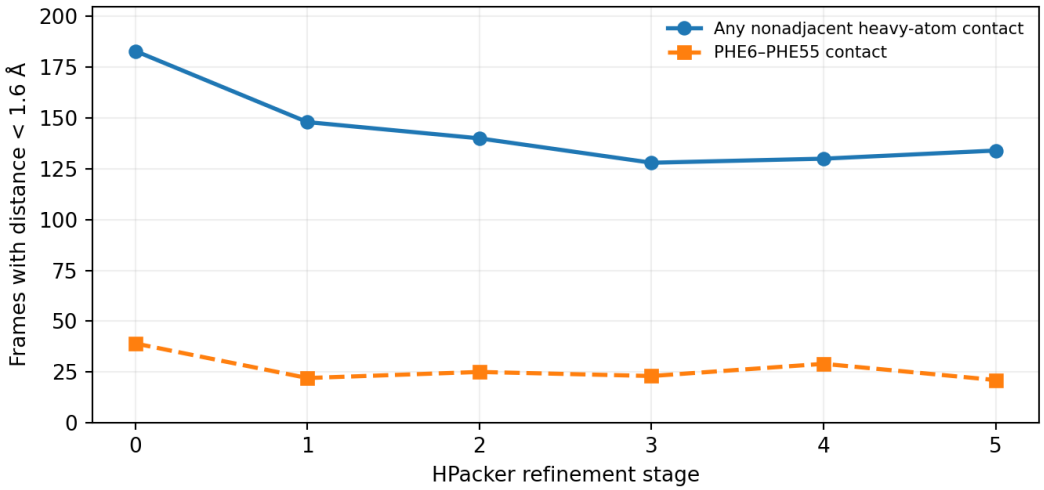


Figure 1. Frame counts with at least one general interresidue heavy-atom contact or a PHE6–PHE55 contact below 1.6 Å, stratified by six HPacker reconstruction stages; common denominator 200.

3.2. Selected WT0 reconstruction history

The minimum PHE6–PHE55 heavy-atom distances in WT0 across stages 0–5 were 4.291, 0.633, 4.037, 0.560, 4.030 and 0.277 Å, respectively.

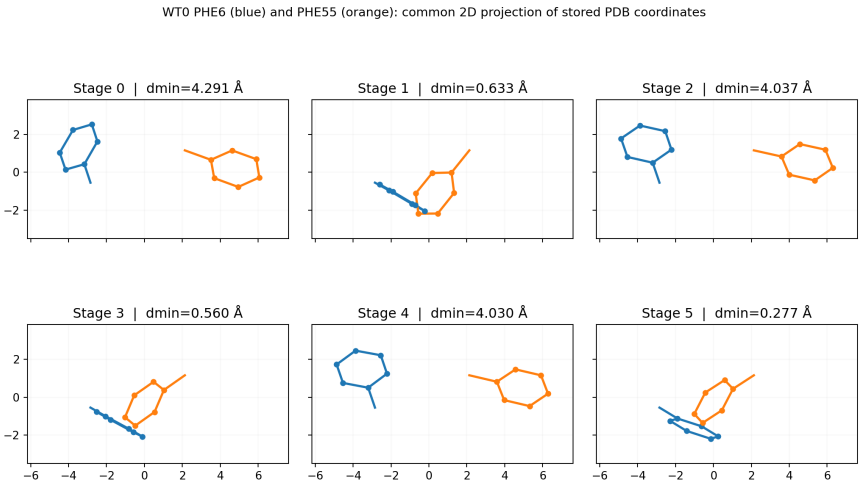


Figure 2. PHE6 (blue) and PHE55 (orange) in WT0 at stages 0–5. Panels use the same planar projection of archived PDB coordinates. Labeled minimum interresidue distances are calculated in three dimensions, not from the drawing.

3.3. Selected intervention and network responses

For WT0 at stage 5, a v8.1 intervention-derived structure increased the minimum PHE6–PHE55 heavy-atom distance from 0.2773 to 1.8815 Å. PHE6–PHE55 atom pairs below 1.6 Å decreased from five to zero, whereas the total protein-wide count decreased from six to one. This selected structure provided the “rescue” input for subsequent comparisons. Separately, post hoc v8.3/v8.4 network diagnostics recorded a WT0 donor 53 → receiver 55 maximum circular χ difference of 54.089° and maximum raw-component difference of 0.1410. Selected WT1 donor 55 → receiver 56, 3 and 59 differences were 178.329°, 148.689° and 52.082°. All six selected low-response examples had nonzero raw-output differences. The proposed WT0 residue-17-associated sequence and WT1 branches are exploratory response relationships, not demonstrated unique causal mediators; see SI S4.

3.4. Selected minimization and relaxation

The selected WT0 baseline and rescue initially had six and one nonadjacent interresidue heavy-atom pairs below 1.6 Å, respectively. Both had zero after minimization. All four selected WT0/WT1 baseline and rescue cases completed initial stabilization, 0.1 ns NVT and 0.4 ns NPT, without nonfinite values at recorded accepted checkpoints; their final archived protein-heavy-atom structures had zero diagnostic close pairs. Rescue was not required for completion in these examples. Total solvated potential energies are not comparable free energies (SI S5).

3.5. Bounded Stage A LJ calculation

The protein heavy-atom LJ proxy produced approximately $+1.662\times10^{13}$ and $+2.707\times10^{4}$ kJ/mol for WT0 baseline and rescue, and $+9.951\times10^{6}$ and $+2.758\times10^{4}$ kJ/mol for WT1. For WT0, PHE6 CD1–PHE55 CE2 changed from 0.277 to 5.514 Å and contributed approximately 99.999999% of the net decrease in this defined proxy. For WT1, PHE6 CG–PHE55 CE2 changed from 0.916 to 6.336 Å and contributed about 99.18%. The values reflect severely nonphysical short contacts and do not measure contributions to the solvated total potential energy.

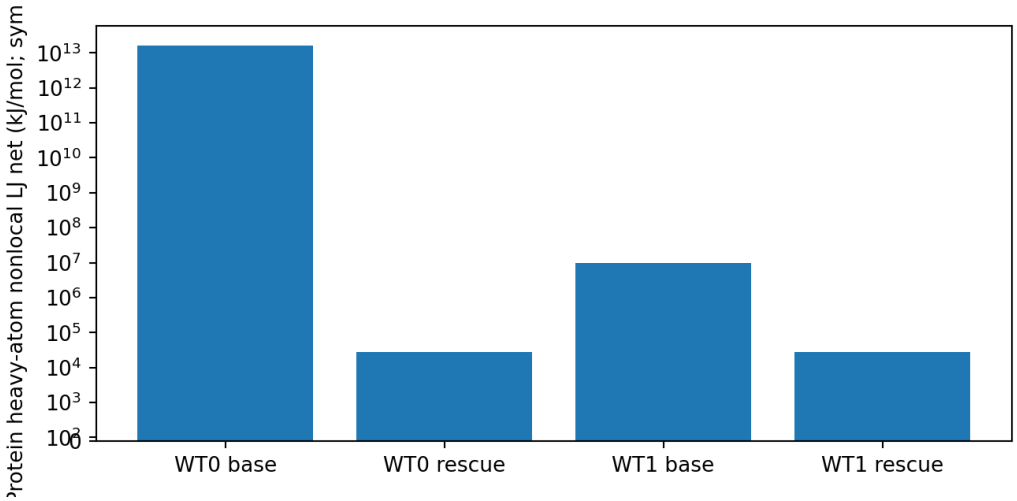


Figure 3. Net protein-only heavy-atom Lennard-Jones proxy values for two selected baseline/rescue pairs, displayed on a logarithmic scale. This is not solvated total potential energy or free energy.

4. Discussion

Saving all refinement stages captures the nonmonotonic presence of extreme atomic overlap: a final-stage snapshot alone obscures the 43 ordered PHE6–PHE55 recontacts and 25 completely alternating frame histories. Selected interventions document local input sensitivity and provide the provenance of structures used in subsequent relaxation and LJ calculations. These findings are descriptive stage-sequence and selected-case observations, not dynamical oscillations or an independently established unique mediation chain. The Stage A calculation quantifies the expected steep repulsive consequence of near-zero atom-pair distances within a deliberately restricted LJ energy definition.

5. Limitations

All 200 source frames derive from one WT ensemble. The PHE pair and local-response examples were selected after exploratory observation, without an independent holdout or prespecified response thresholds. Original NN parity and sham controls are incompletely authenticated; Phase 2B was not independently re-audited. Four selected relaxation runs cannot support general MD-success comparisons. Stage A omits solvent, ion, electrostatic, bonded and hydrogen-pair LJ terms; no whole-system energy attribution or equilibrium free-energy claim is supported. Archived backbone coordinates were unchanged at recorded PDB precision, not verified bitwise at all internal stages.

6. Conclusions

In this archived BioEmu-to-HPacker case, extreme heavy-atom proximity was common and some selected PHE6–PHE55 contacts disappeared and returned during sequential reconstruction. Local intervention and NN-response records, together with bounded minimization, relaxation and LJ diagnostics, provide a traceable description of selected cases. The record does not establish a generally superior side-chain packing method, thermodynamic stabilization or improvement in MD outcomes.

7. Data availability and declarations

Companion release files include 200-frame/1,200-stage diagnostic tables, selected PDBs, archived v8.1–v8.4 intervention implementations, selected response summaries, continuation traces, Stage A LJ data, verification scripts and checksums. Original software and model weights remain subject to upstream licenses. Author: Sho Miyazaki (Independent researcher, Tokyo, Japan). Funding: no external funding declared for this independent analysis. Competing interests: none declared. Acknowledgments: BioEmu, HPacker and OpenMM developers are credited through citation; no additional personal acknowledgments are claimed. AI disclosure: ChatGPT assisted with study planning, code and data checks, figures, drafting and QC; the human author remains responsible for the scientific interpretation and release. Author-owned prose and figures: CC BY 4.0; owned original code: MIT; see license scope for third-party materials. A repository URL and DOI will be populated upon deposition.

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